

EXCLUSIVE

IAS PRELIMS 2026

CURRENT AFFAIRS

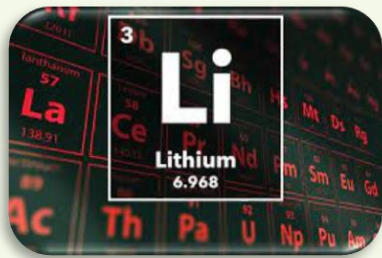
REVISION

GS-1 (DYNAMIC)

MODULE - 1



**SCIENCE
&
TECHNOLOGY-1**



- It was first recognized as an element in 1817 when Swedish chemist Johan Arfwedson analyzed the mineral petalite.
- With a specific gravity of 0.534, it is about half as dense as water and the lightest of all metals.
- In its pure elemental form, it is a soft silvery white metal, but it is highly reactive and therefore never is found as a metal in nature.
- It has an average concentration of 20 PPM in the earth's continental crust.
- It is more abundant than some of the better known metals including Tin and silver.

It is a soft light metal that has many uses including making lightweight rechargeable batteries for computers, cell phones, other digital devices and electric cars. It occurs in most rocks as a trace element.



LITHIUM DID NOT NATURALLY FORM ON THE PLANET!

- Previously, 1,600 tonnes of lithium reserves were found in Karnataka's Mandya district. However, it was not commercially viable.
- Lithium did not naturally form on the planet. Scientists have suggested that this is a cosmic element that formed from the bright stellar explosions called Novae.

According to the World Bank, mining of crucial minerals will need to increase by 500% to meet global climate targets by 2050.

- A NASA funded study found that while the Big Bang created a small amount of lithium in the initial formation of the universe, the majority of lithium gets manufactured in the nuclear reactions that power the Novae explosions distributing the mineral throughout the Galaxy.
- The 2019 Nobel Prize in chemistry was awarded to the scientists, who worked on the lithium-ion battery. Till now, India has depended on Australia and Argentina for lithium imports.



LITHIUM HAS MULTITUDE OF APPLICATIONS!



- Lithium is added to glasses and ceramics for strength and resistance to temperature change. It is used in heat resistant greases and lubricants.
- It is alloyed with aluminium and copper to save weight in air frame structural components.
- Lithium is used in certain psychiatric medications and in dental ceramics.
- The lighter of the two lithium isotopes ${}^6\text{Li}$ is used in the production of tritium for nuclear weapons.
- In 2015, less than 30% of lithium demand was for batteries. The bulk of demand was split between ceramics and glasses (35%) and the greases, metallurgical powders, polymers and other industrial uses (35+%).
- By 2030, batteries are expected to account for 95% of the total lithium demand.

WHY ARE LITHIUM-ION BATTERIES USED IN MOST ELECTRIC VEHICLES?

Learning Space



Research and Development is ongoing to reduce the relatively high cost, extend their useful life and address the safety concerns in regard to overheating.

- Different types of batteries are available in the market, which include lithium-ion batteries, solid-state batteries, nickel metal hydride batteries and lead acid batteries etc.
- Among the above, lithium-ion batteries are used in the electric vehicles because of their high energy per unit mass in comparison to other electrical energy storage systems.
- They also have a high power to weight ratio, high energy efficiency, good high temperature performance and low self discharge.
- Most components of lithium-ion batteries can be recycled. In view of the above, these are used in a variety of ways from electric vehicles to residential batteries to grid-scale applications.
- Though lithium-ion can refer to a wide array of chemistries, it ultimately consists of a battery based on charge and discharge reactions from a lithiated metal oxide cathode and a graphite anode.

Technology	Description	Maturity	Lithium recovery, %
Adsorbents	Adsorption process using sorbents	In use commercially	80–99.9
Ion exchange	Ion exchanger using resins, aluminates, or ceramics	Precommercial	80–99.9
Solvent extraction	Fluid solvent mixture blended with brine to extract water	Precommercial	99.9

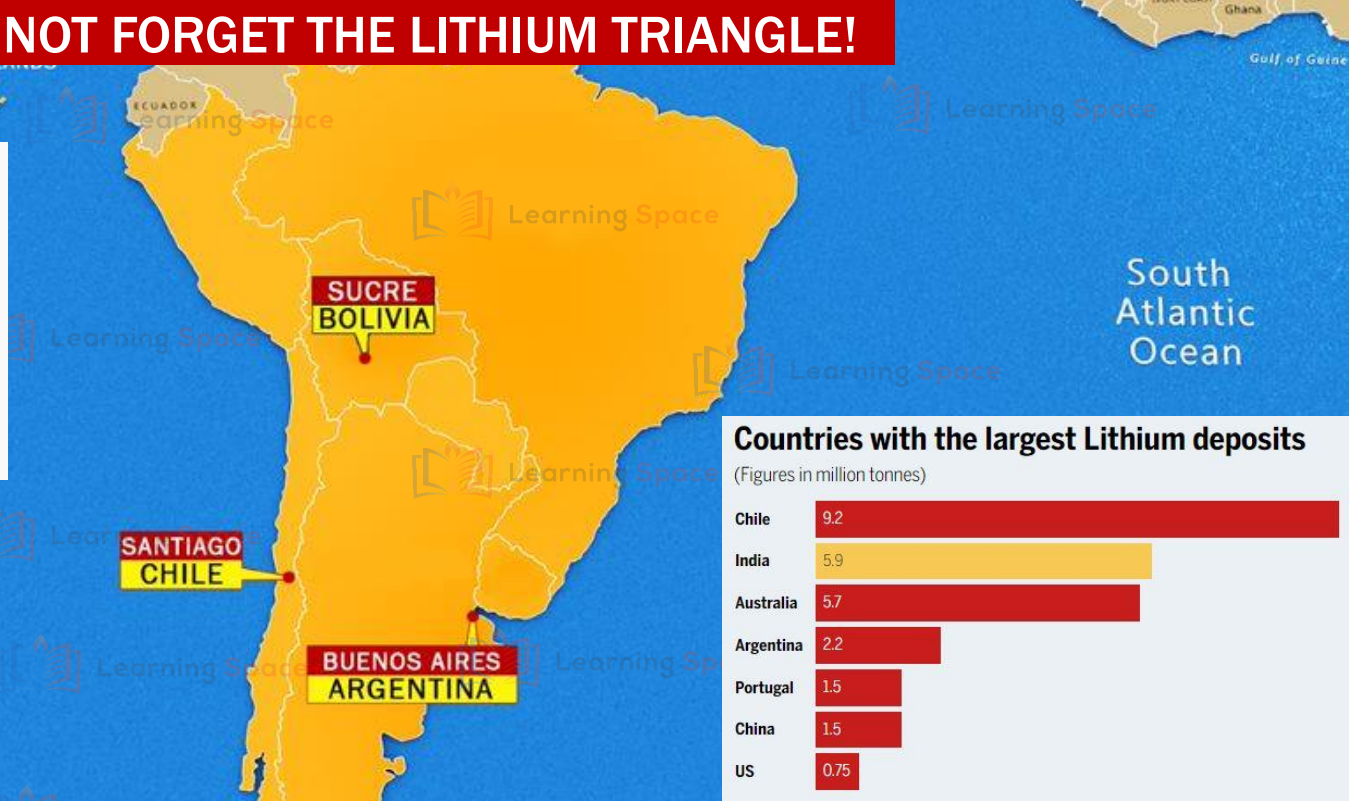
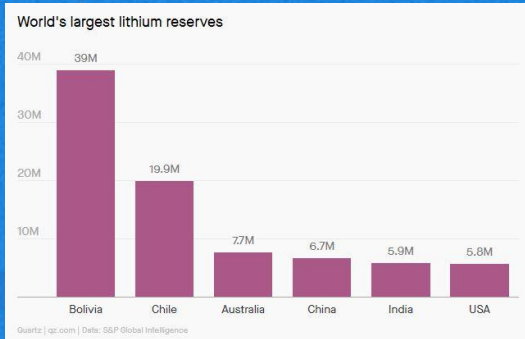
Lithium is also used to treat bipolar disorder, helping to stabilize wild mood swings caused by the illness.

THE FINDINGS ARE BASED ON THE PRELIMINARY EXPLORATION (G3)

- The G3 advanced study of GSI shows the presence of best quality lithium in abundant quantity in the foothills of Mata Vaishno Devi shrine at salal village (Reasi district) of Jammu and Kashmir.
- Against the normal grade of 220 PPM, the lithium found in J&K is of 500 PPM + grading. With a stock pile of 5.9 million, India will surpass China in its availability.
- This exploration is called G3 stage and as per the UNFC classification, there are four stages of exploration for any mineral deposit
 - ✓ Reconnaissance (G4)
 - ✓ Preliminary exploration (G3)
 - ✓ General exploration (G2)
 - ✓ Detailed exploration (G1)

The word lithium comes from “lithos”, the Greek word for “stone”.

DO NOT FORGET THE LITHIUM TRIANGLE!



- The vast majority of the world's lithium reserves are found in the wide expanse of salt flats in Bolivia, Chile and Argentina.
- Bolivia has by far the world's largest lithium reserves, but most of it remains in the ground as the country does not have the developed infrastructure to excavate it. Due to this, Bolivia does not rank anywhere near the top in actual lithium production

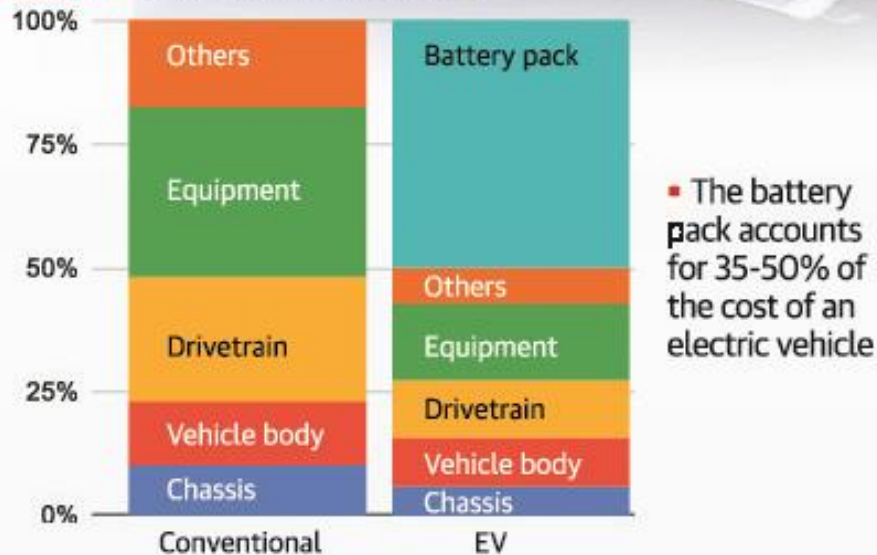
THERE ARE ALSO ASSOCIATED ENVIRONMENTAL CONCERNS!



- The process of mining lithium is not environment friendly. Lithium is extracted from hard rocks and underground brine reservoirs largely found in Australia, Chile and Argentina.
- After it is mined, it is roasted using fossil fuels, searing the landscape and leaving behind scars. The extraction process also requires a lot of water and releases large amounts of carbon dioxide into the atmosphere.

BATTERY PACK ACCOUNTS FOR UP TO 50% OF COST OF ELECTRIC VEHICLE

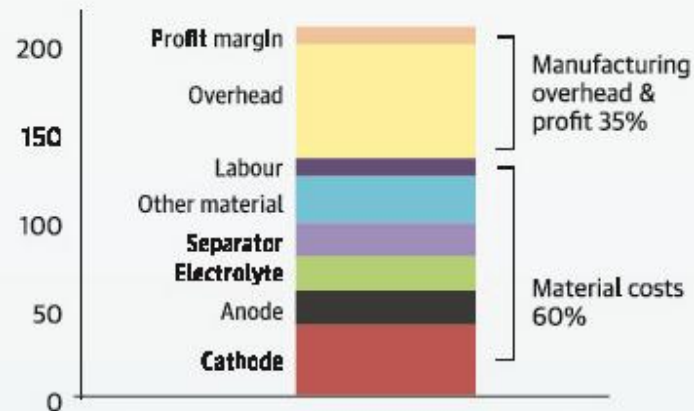
Chart 1: Overall cost break-up



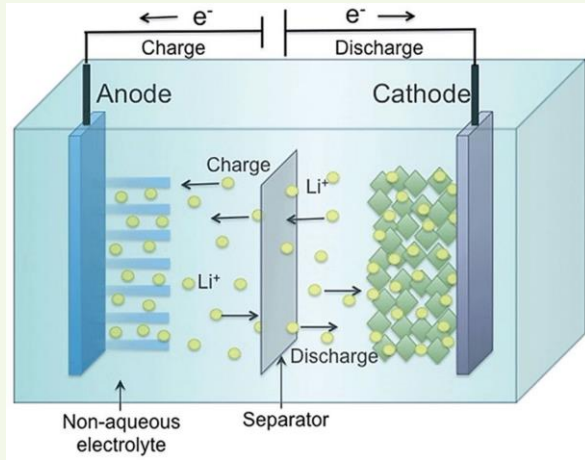
SEE THE BREAK-UP OF VARIOUS COMPONENTS

Chart 3: Lithium-ion battery cost break-up

- Raw material prices alone contribute 60% of the battery costs



Source: Qnovo, an organisation based in the U.S. that works in the field of batteries



- A battery is a combination of cells.
- In a cell, there are two electrodes — the anode, which is the electron giver, and the cathode, which is the electron acceptor.
- The electrons move from the anode to the cathode via an external circuit - that 'flow' is electricity.
- The atoms that have lost electrons, to become positively charged ions, also move from the anode to the cathode, but through the electrolyte.
- The anode in Lithium-ion battery is typically made up of Graphite, coated on Copper Foil.
- The battery generates electricity through chemical reactions of lithium. However, since lithium is unstable in the element form, the combination of lithium and oxygen, lithium oxide is used. The Cathode determines the capacity and voltage of a Li-ion battery.

EVs NEED MORE MINERALS THAN A CONVENTIONAL VEHICLE

EVs need more minerals than a conventional vehicle

	Conventional vehicle	Electric vehicle
Graphite	-	66.3
Nickel	-	39.9
Copper	22.3	53.2
Lithium	-	8.9
Cobalt	-	13.3
Manganese	11.2	24.5



- The **anode** in **Li-ion batteries** requires **graphite** and, apparently, there are no substitutes for it.
- The **cathodes**, which also require **Li-ion**, contains **nickel** to deliver high energy density, allowing the vehicle to travel farther.

EXPECTED MINERAL DEMAND OF EV SECTOR BY 2030

Expected mineral demand of the EV sector
(in thousand tonnes)

	2020	2030	Growth (%)
Graphite	141.03	2,499.25	1,672
Nickel	80.47	1,566.94	1,847
Copper	110.32	1,632.63	1,380
Lithium	19.83	358.39	1,707
Cobalt	21.12	256.64	1,115
Manganese	25.34	246.28	872

Source: UK, British Geological Survey, IEA, India Exim Bank Research,

Cobalt is important for **Evs**, as it prevents the **cathode** from **overheating**, while extending the **life of batteries**.

FOR LITHIUM, COBALT AND NICKEL ... OUR IMPORT DEPENDENCE IS 100%!

Top producers of rare minerals 2020

Metal	Share of key producers	India's production share
Lithium	Australia (49%), Chile (22%), China (17%), Argentina (8%), Brazil (2%)	-
Graphite	China (65%), Brazil (10%), Madagascar (5%), India (3%), Mozambique (2%)	3%
Cobalt	Congo (68%), Australia (4%), Cuba (4%), Canada (3%), Philippines (3%)	-
Manganese	South Africa (32%), Gabon (16%), China (13%), Australia (10%)	5.32%
Nickel	Indonesia (33%), Philippines (13%), Russia (9%), New Caledonia (8%) Australia (7%)	-

In 2020, Australia was responsible for 49% of global lithium production, while 65% of graphite is produced in China, 68% of cobalt in Congo, and 33% of Nickel in Indonesia.



Fight against disease

THE DISEASE

India accounts for a fifth of global cervical cancer cases with 1.23 lakh cases and around 67,000 deaths every year, according to WHO's International Agency for Research on Cancer.

Cervical cancer kills one woman in every eight minutes in India

THE VACCINE

■ Serum Institute of India's (SII) Cervavac is based on VLP (virus-like particles), similar to the Hepatitis B vaccine

■ It protects by generating antibodies against the HPV virus L1 protein.

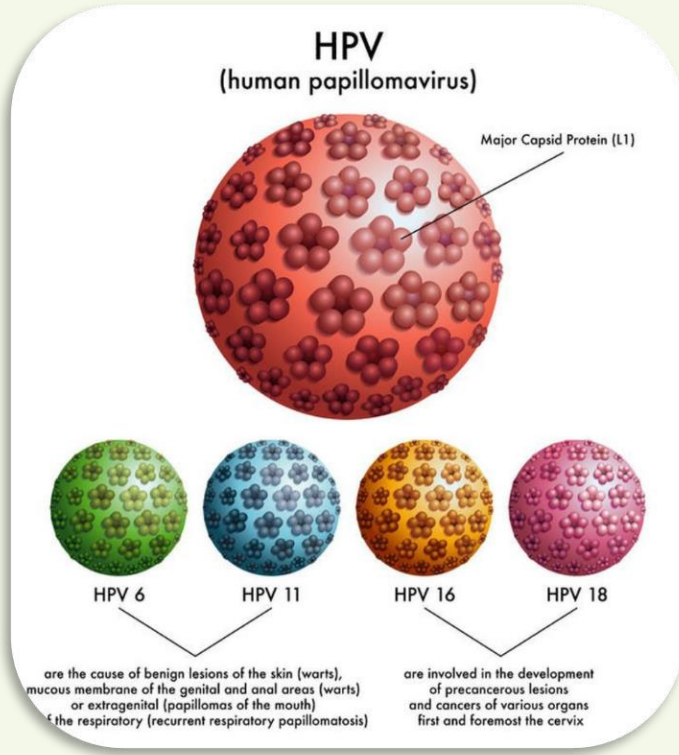
■ Cervavac is expected to cost much less than two vaccines licensed globally which are currently available in India - Gardasil from Merck and Cervarix from GlaxoSmithKline.

■ SII received approval for market authorisation on July 12

HUMAN PAPILLOMAVIRUS (HPV) AND CERVICAL CANCER

- Human papillomavirus (HPV) is the name for common group of viruses and there are more than 100 types of HPV.
- Just like Rota virus, Polio virus, HPV is an example of non-enveloped virus.
- Cervical cancer, the second most common cancer in females after breast cancer, is a major threat faced by women in India.
- HPV virus is found in more than 90% of cervical cancers. Hence, HPV virus is considered to be the most important risk factor for cervical cancer.
- As per Lancet, more than 58% of all cases of cervical cancer and deaths globally were estimated in Asia and India accounts for 21% of cases and 23% of deaths.

There are vaccines which protect against carcinogenic HPV.



- Certain types of the HPV virus can transform normal cervical cells into abnormal ones, and over a long period, some of these abnormal cells may then develop into cervical cancer.
- More than 70% of cervical cancer cases can be attributed to two types of the virus, HPV-16 and HPV-18.
- Vaccination against HPV is the best way to prevent cervical cancer. Ideally, the vaccination should be administered at an early age, before sexual contact.
- Nevertheless, all women should take the vaccine, as it is not necessary that they would have contracted this virus only through sexual activity.

The HPV vaccine covers the age group of nine to 26 and can be given up to 45 years of age, if not adequately vaccinated when one was younger.

CENTRAL GOVERNMENT ANNOUNCED VACCINATION PROGRAMME AGAINST CERVICAL CANCER

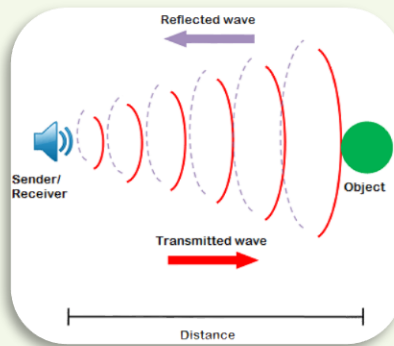
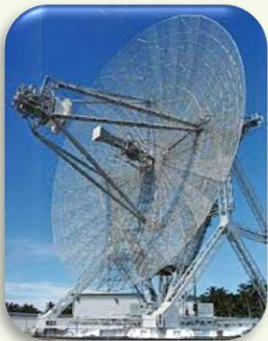
- Central Government announced that it will roll out vaccination for girls aged between 9 and 14 years through schools.
- The Government also clarified that girls who do not go to school will be reached through community outreach and mobile teams.
- India is expected to roll out the indigenously developed Cervavac vaccine by mid-2023 and It has already received the DCGI's approval. It has been cleared by the National Technical Advisory Group for Immunisation for use in the UIP.

- The WHO has specified that countries must reach and maintain an incidence rate of fewer than 4 new cases of cervical cancer per 1,00,000 women a year. To achieve that goal, it is necessary that 90% of girls will have to be fully vaccinated with the HPV vaccine by the age of 15.



WHAT DO YOU UNDERSTAND BY A RADAR?

- **RADAR** is the expansion for **Radio, Detection and Ranging**. It works on the principle of **electromagnetic waves sent out by the transmitter**.
- The same wave that **strikes an object** is **reflected back** to the **receiver**. The **distance** up to the object is **determined** based on the **speed** of the electromagnetic wave, and the **time to travel** to the **object and back**.



KNOW ABOUT THESE SPECIAL RADARS!

- The **Ground Penetrating Radar** studies the Earth's crust up to **9-metres in depth**. It is being used by the Defence Geoinformatics Research Establishment (**DGRE**) at **Joshimath**.
- The Interferometric Synthetic Aperture Radar (**InSAR**), that makes **high density measurements** over large areas by using radar signals from **Earth orbiting satellites** and measures changes in land-surface, is also being used in **Joshimath** and other parts of **Uttarakhand**.

DOPLER RADAR IS BASED ON THE DOPPLER PRINCIPLE!

- A Doppler Radar is a specialised radar that uses the Doppler effect to produce velocity data about objects at a distance.
- When the source and the signal are in relative motion to each other, there is a change in the frequency observed by the observer. This is called the Doppler effect. If they are moving closer, the frequency increases and vice versa. To conclude, a Doppler Weather Radar (DWR) works on the Doppler principle.

DWR has the equipment to measure rainfall intensity, wind shear and velocity and locate a storm center and the direction of a tornado or gust front.

TO SUMMARIZE ... DWRs FACILITATE PRECISE DETECTION OF VARIOUS WEATHER ASPECTS!

- DWRs give the most precise detection of weather parameters, which include
 - ✓ dynamic weather events such as turbulence, cyclones, thunderstorms, lightning etc.
 - ✓ volumetric analysis of clouds that help in the quantification of rain forecasts
 - ✓ cyclonic intensity
 - ✓ Precipitation etc.

DWRs precisely detect normal or routine events in real time too! A DWR costs approximately Rs. 10-20 Crores per unit. Operation and maintenance cost, depending on type, is generally Rs. 20 - 40 lakh per annum.

Indian cough syrup market is set to soar from around \$262 million in 2024 to \$743 million by 2035.



SEE THE COUGH SYRUP DEATHS DURING THE PAST FIVE YEARS!

- Diethylene glycol in Indian-made cough syrups has claimed dozens of young lives.
- Between December 2019 and January 2020, at least 12 children under five years died in Jammu and Kashmir allegedly from cough syrup.
- In 2023, Indian syrups tainted with diethylene glycol were linked to the deaths of 70 children in the Gambia and 18 in Uzbekistan.

There is a weak regulatory system struggling to oversee hundreds of low-cost, often unapproved syrups produced by smaller manufacturers and sold over the counter.

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SINCERE THANKS
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